

MQTT – Modular Controller Board

Summary:

ESP32 S3 based controller for attached modules. Designed to receive commands via MQTT and send required information. This version supports four individual modules being attached. USB C connection for programming, and only supports the controller itself.

Table of Contents

Page 2: ESP32 S3 WROOM 1U
Page 3: Power Input
Page 4: Power Conversion
Page 5: Module Connectors
Page 6: USB Input

Non BOM Items

- 

H101
MountingHole 3.5mm
- 

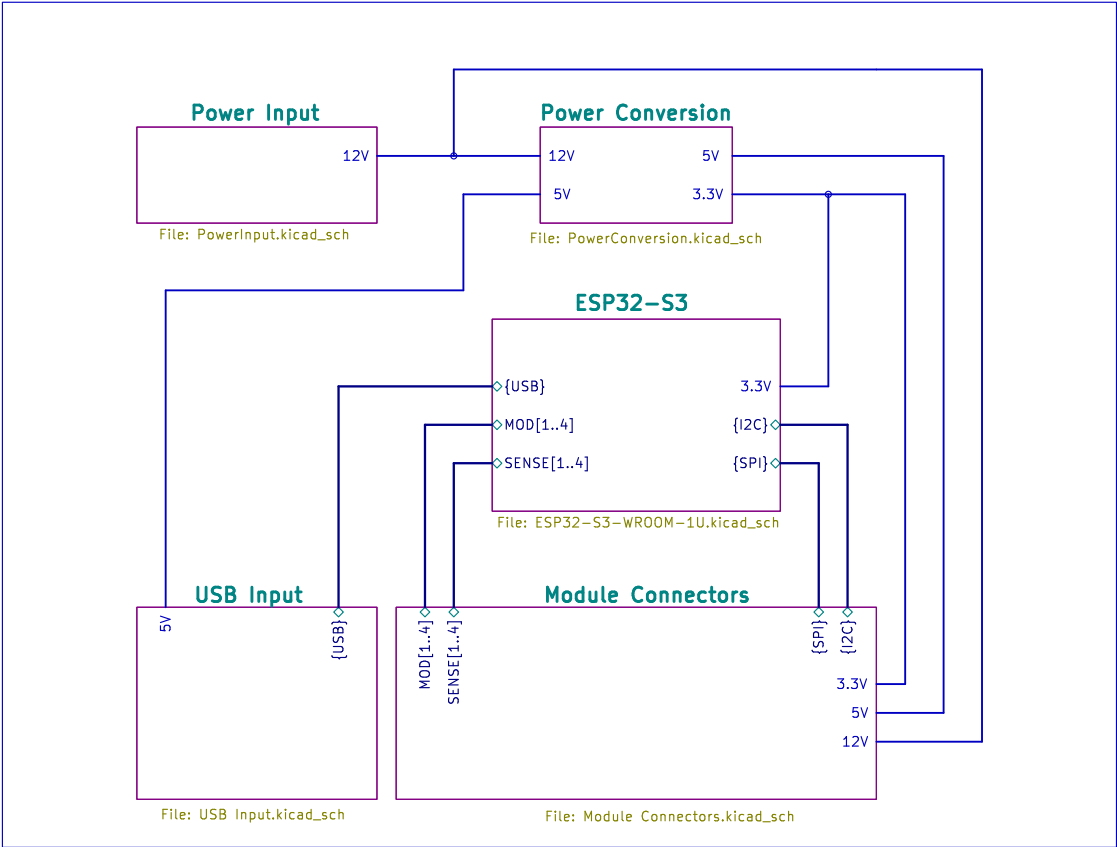
H103
MountingHole 3.5mm
- 

H102
MountingHole 3.5mm
- 

H104
MountingHole 3.5mm

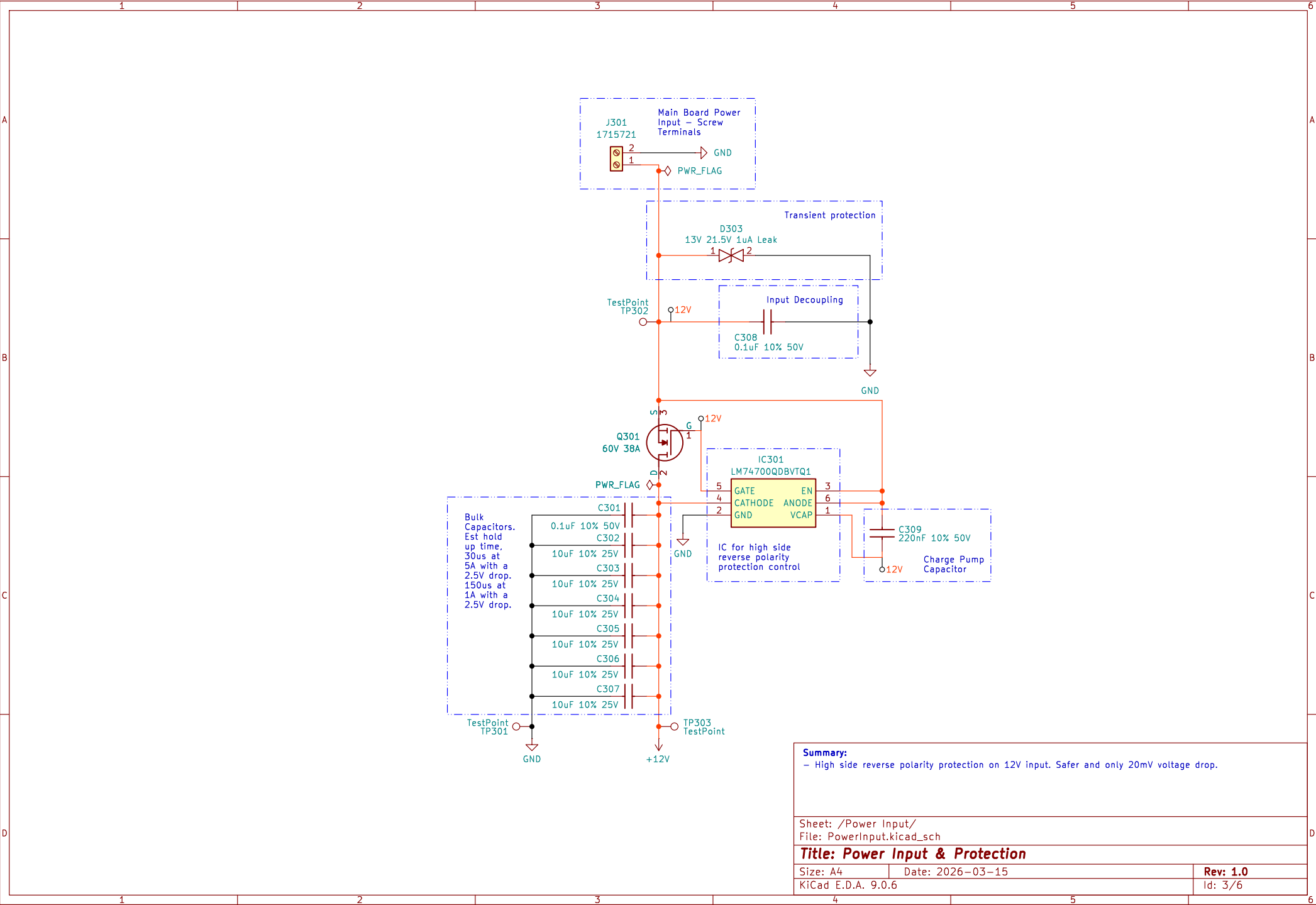
Additional BOM Items

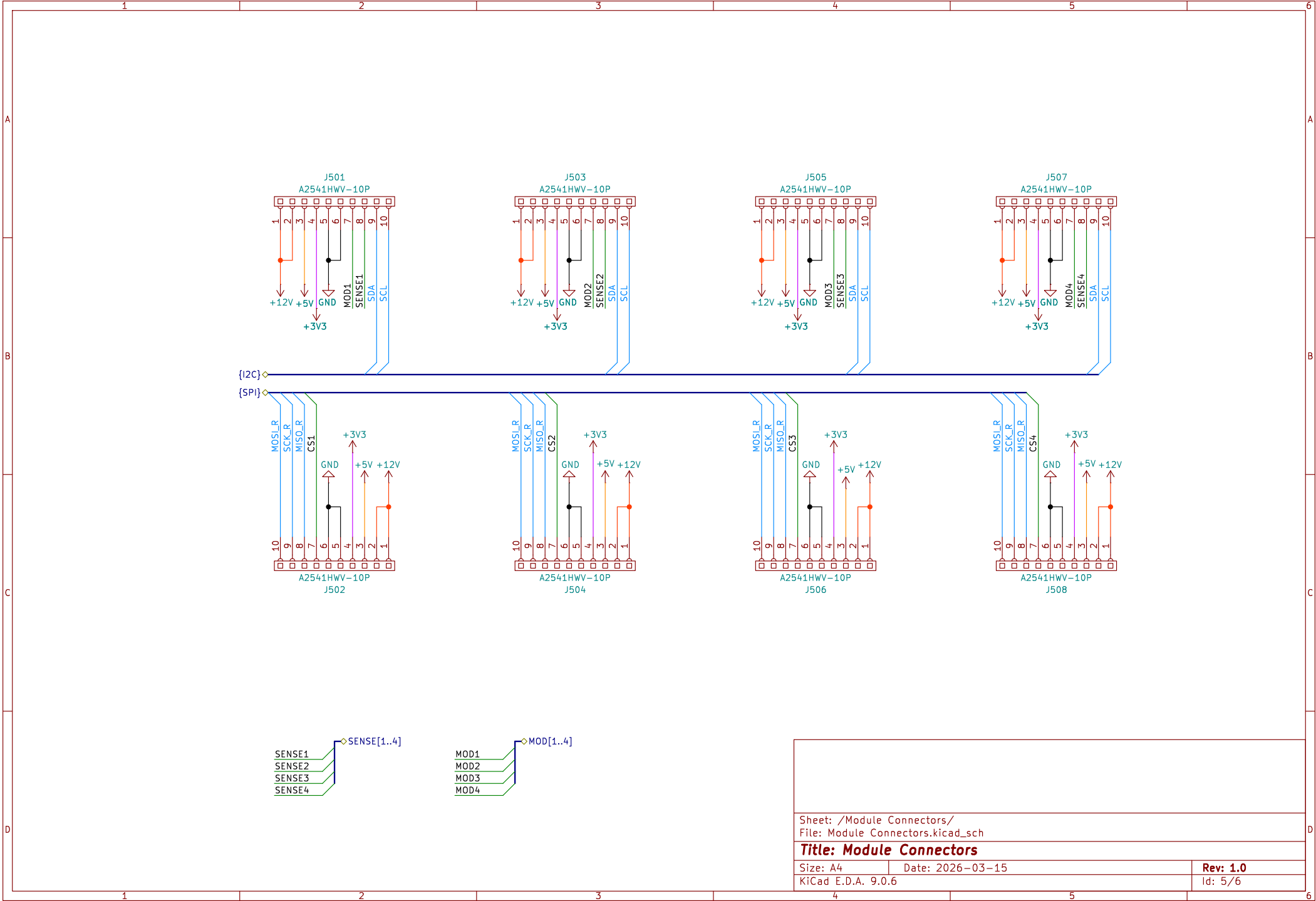
—

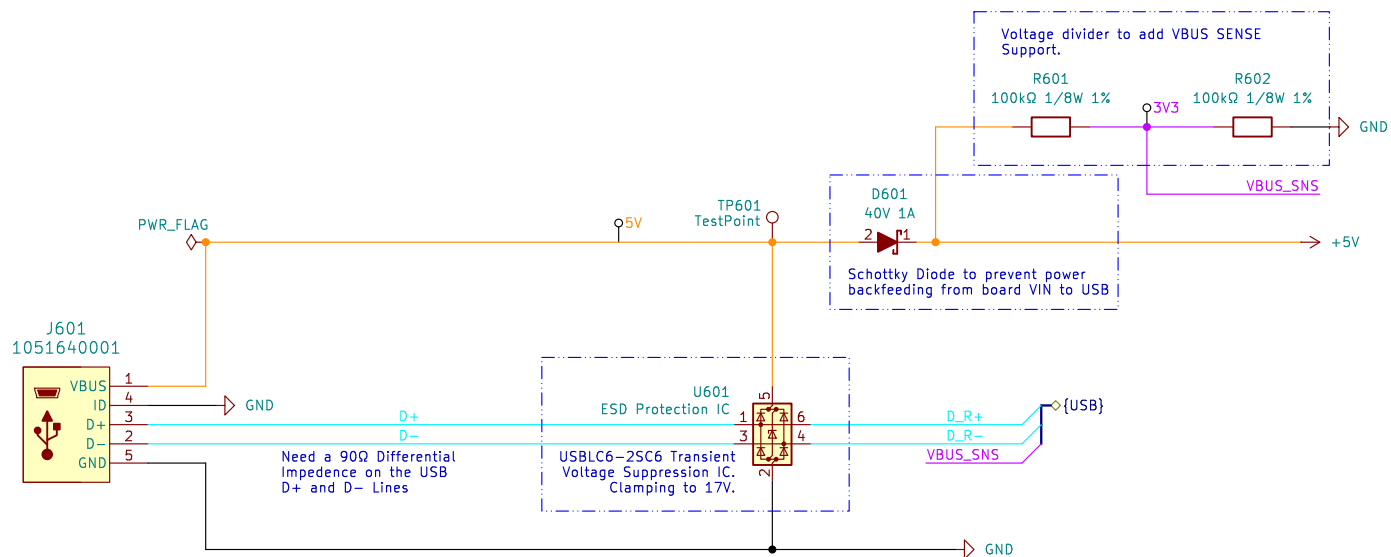


Sheet: /		
File: MQTT-ModularControllerBoard.kicad_sch		
Title: MQTT – Modular Controller Board		
Size: A4	Date: 2026-03-14	Rev: 1.0
KiCad E.D.A. 9.0.6	Id: 1/6	









Summary:
- VBUS_SNS in place for USB connection while connected to 12V. Requires detection of being self powered on startup in software.

Sheet: /USB Input/
File: USB Input.kicad_sch

Title: USB C Connection

Size: A4 Date: 2026-03-18

KiCad E.D.A. 9.0.6

Rev: 1.0

Id: 6/6